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Gokulakkannan P
Generative AI
ex.no:08

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```python
# ==============================
# 1. Install Dependencies
# ==============================
!pip install streamlit transformers rouge-score nltk
!npm install -g localtunnel

import nltk
nltk.download('punkt')

# ==============================
# 2. Create Streamlit App
# ==============================
app_code = """
import streamlit as st
from transformers import pipeline
from rouge_score import rouge_scorer
from nltk.translate.bleu_score import sentence_bleu

# Load model
@st.cache_resource
def load_model():
    return pipeline("text2text-generation", model="google/flan-t5-base")

qa_pipeline = load_model()

# UI
st.title("📄 AI Customer Support / QA System")
```

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```python
reference = st.text_area("Enter reference answer (for evaluation)")

if st.button("Generate Answer"):
    if query:
        response = qa_pipeline(query, max_length=150)[0]['generated_text']

        st.subheader("🤖 Generated Answer")
        st.write(response)

        # Evaluation
        if reference:
            scorer = rouge_scorer.RougeScorer(['rouge1', 'rougeL'], use_stemmer=True)
            rouge_scores = scorer.score(reference, response)

            bleu = sentence_bleu([reference.split()], response.split())

            st.subheader("📊 Evaluation Metrics")
            st.write("ROUGE-1:", round(rouge_scores['rouge1'].fmeasure, 4))
            st.write("ROUGE-L:", round(rouge_scores['rougeL'].fmeasure, 4))
            st.write("BLEU Score:", round(bleu, 4))
"""

# Save file
with open("app.py", "w") as f:
    f.write(app_code)

# ==============================
# 3. Run Streamlit + LocalTunnel
# ==============================
import subprocess, time
```

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```
# Wait for server to start
time.sleep(10)

# Start LocalTunnel
lt_process = subprocess.Popen(
    ["lt", "--port", "8501"],
    stdout=subprocess.PIPE,
    stderr=subprocess.PIPE,
    text=True
)

# Print public URL
for line in lt_process.stdout:
    if "your url is" in line.lower():
        print("🚀 Open this link in browser:", line.strip())
        break
```

```
Requirement already satisfied: protobuf<8,>=3.20 in /usr/local/lib/python3.12/dist-packages (from streamlit) (5.29.6)
Requirement already satisfied: pyarrow>=7.0 in /usr/local/lib/python3.12/dist-packages (from streamlit) (18.1.0)
Requirement already satisfied: requests<3,>=2.27 in /usr/local/lib/python3.12/dist-packages (from streamlit) (2.32.4)
Requirement already satisfied: tenacity<10,>=8.1.0 in /usr/local/lib/python3.12/dist-packages (from streamlit) (9.1.4)
Requirement already satisfied: toml<2,>=0.10.1 in /usr/local/lib/python3.12/dist-packages (from streamlit) (0.10.2)
Requirement already satisfied: tornado!=6.5.0,<7,>=6.0.3 in /usr/local/lib/python3.12/dist-packages (from streamlit) (6.5.1)
Requirement already satisfied: typing-extensions<5,>=4.10.0 in /usr/local/lib/python3.12/dist-packages (from streamlit) (4.15.0)
Requirement already satisfied: watchdog<7,>=2.1.5 in /usr/local/lib/python3.12/dist-packages (from streamlit) (6.0.0)
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from transformers) (3.29.0)
Requirement already satisfied: huggingface-hub<2.0,>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (1.11.0)
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-packages (from transformers) (6.0.3)
Requirement already satisfied: regex!=2019.12.17 in /usr/local/lib/python3.12/dist-packages (from transformers) (2025.11.3)
```

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```
Requirement already satisfied: smmap<6,>=3.0.1 in /usr/local/lib/python3.12/dist-packages (from gitdb<5,>=4.0.1->gitpython!=3.1.19,<4,>=3.0.7->streamli
Requirement already satisfied: anyio in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.3.0->transformers) (4.15.
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.3.0->transformers) (1.0.9)
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*->httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.3.0->transformers) (0.16.
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from jinja2->altair!=5.4.0,!=5.4.1,<7,>=4.0->streamlit) (3.0.3)
Requirement already satisfied: attrs>=22.2.0 in /usr/local/lib/python3.12/dist-packages (from jsonschema>=3.0->altair!=5.4.0,!=5.4.1,<7,>=4.0->streamlit) (26.1.0)
Requirement already satisfied: jsonschema-specifications>=2023.03.6 in /usr/local/lib/python3.12/dist-packages (from jsonschema>=3.0->altair!=5.4.0,!=5.4.1,<7,>=4.0->streami
Requirement already satisfied: referencing>=0.28.4 in /usr/local/lib/python3.12/dist-packages (from jsonschema>=3.0->altair!=5.4.0,!=5.4.1,<7,>=4.0->streamlit) (0.37.0)
Requirement already satisfied: rpds-py>=0.25.0 in /usr/local/lib/python3.12/dist-packages (from jsonschema>=3.0->altair!=5.4.0,!=5.4.1,<7,>=4.0->streamlit) (0.30.0)
Requirement already satisfied: shellingham>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0,>=1.3.0->transformers) (1.5.4)
Requirement already satisfied: rich>=12.3.0 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0,>=1.3.0->transformers) (13.9.4)
Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0,>=1.3.0->transformers) (0.0.4)
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer->huggingface-hub<2.0,>=1.3.0->transformers) (4.0.0
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer->huggingface-hub<2.0,>=1.3.0->transformers) (2.2
Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0->rich>=12.3.0->typer->huggingface-hub<2.0,>=1.3.0->transfor
added 22 packages in 3s

3 packages are looking for funding
  run `npm fund` for details
[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data]   Package punkt is already up-to-date!
🚀 Open this link in browser: your url is: https://empty-trains-post.loca.lt
```